Python Introduction (Detailed Guide for PDF)

What is Python?

Python is a high-level, interpreted, and general-purpose programming language known for its simple and readable syntax.

It is widely used in:

- Web development
- Data science
- Artificial intelligence
- Automation
- Software development

History of Python

- Created by Guido van Rossum
- Released in 1991
- Named after "Monty Python"

Why Use Python?

- Easy to learn and use
- Large community support
- Extensive libraries
- Cross-platform compatibility

Features of Python

- Simple syntax
- Interpreted language
- Object-oriented
- Dynamically typed
- Extensive libraries

Python Syntax (Basic Example)

```
print("Hello, World!")
```

Variables

```
x = 10
y = "Hello"
```

- No need to declare type

Data Types

- int
- float
- str
- bool
- list
- tuple
- dictionary

Example:

```
name = "John"
age = 25
```

Operators

- Arithmetic (+, -, *, /)
- Comparison (==, !=)
- Logical (and, or, not)

Conditional Statements

```
age = 20

if age > 18:
```

```
    print("Adult")
else:
    print("Minor")
```

Loops

For Loop

```
for i in range(5):
    print(i)
```

While Loop

```
i = 0
while i < 5:
    i += 1
```

Functions

```
def greet():
    print("Hello")

greet()
```

Lists

```
numbers = [1, 2, 3]
```

Tuples

```
numbers = (1, 2, 3)
```

Dictionaries

```
person = {
    "name": "John",
    "age": 25
}
```

String Handling

```
text = "hello"
print(text.upper())
```

File Handling

```
with open("file.txt", "r") as file:
    data = file.read()
```

Exception Handling

```
try:
    x = 10 / 0
except:
    print("Error")
```

Object-Oriented Programming

```
class Person:
    def __init__(self, name):
        self.name = name
```

Modules and Packages

```
import math
print(math.sqrt(16))
```

Example Project

```
name = input("Enter your name: ")
print("Hello", name)
```

Advantages of Python

- Easy to learn
- Large libraries
- Versatile

Disadvantages of Python

- Slower than compiled languages
- High memory usage

Best Practices

- Use meaningful variable names
- Follow indentation rules
- Write clean and readable code